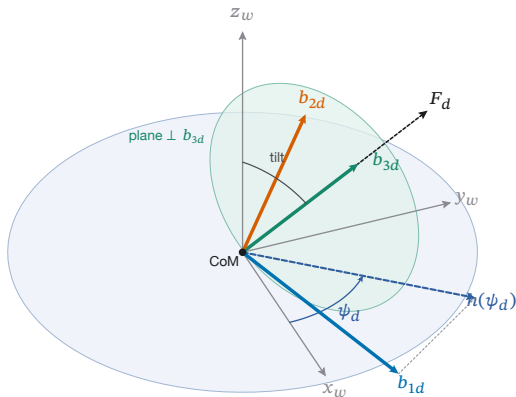


a Desired attitude from thrust direction and heading `core/rotations.py`

$$\begin{aligned} b_{3d} &= F_d / \|F_d\| \\ b_{2d} &= \frac{b_{3d} \times h}{\|b_{3d} \times h\|} \\ b_{1d} &= b_{2d} \times b_{3d} \\ h(\psi_d) &= (\cos \psi_d, \sin \psi_d, 0) \end{aligned}$$



$R_d = [b_{1d} \ b_{2d} \ b_{3d}]$: body +z follows the thrust; b_{1d} is the heading h projected onto the plane $\perp b_{3d}$. ψ_d is an auxiliary heading, not the Z-Y-X yaw of a tilted body.
 Degenerate $b_{3d} \parallel h$ or $F_d = 0$: hold the previous axis / attitude (no requested yaw rates).

d How jerk and snap reach the attitude loop

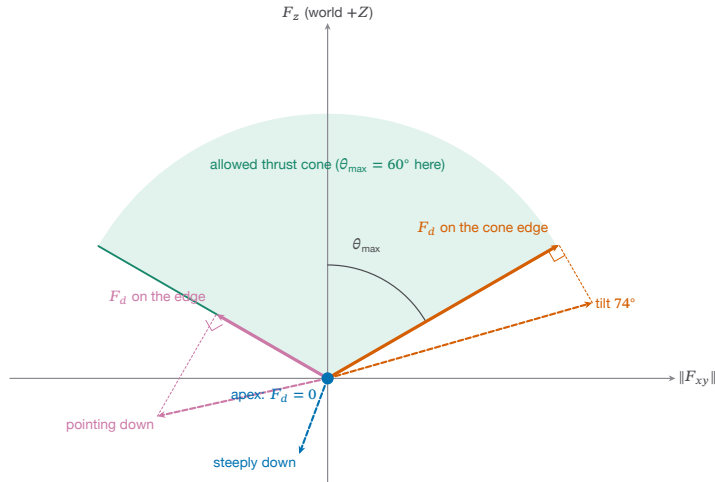
$$b_{3d} = \frac{F_d}{\|F_d\|}, \quad \dot{b}_{3d} = \frac{\dot{F}_d - b_{3d} (b_{3d}^T \dot{F}_d)}{\|F_d\|}, \quad \ddot{b}_{3d} \text{ likewise from } \ddot{F}_d$$

$$b_{2d}, b_{1d} \text{ and their derivatives from } \dot{h} = \dot{\psi}_d h^\perp, \quad \ddot{h} = \ddot{\psi}_d h^\perp - \dot{\psi}_d^2 h$$

$$\hat{\Omega}_d = R_d^T \dot{R}_d, \quad \hat{\Omega}_d = \dot{R}_d^T \dot{R}_d + R_d^T \ddot{R}_d \text{ (vee of the skew part)}$$

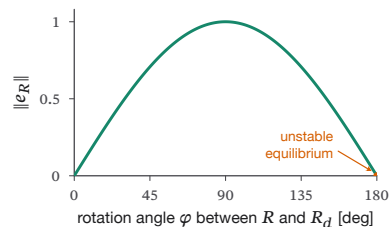
With $\dot{F}_d = m \dot{j}_d + \dot{F}_{ib}$ and $\dot{F}_d = m s_d + \dot{F}_{ib}$ the feedforward rates are analytic for the reference part.
 The cascaded PID passes $\dot{F}_d = \ddot{F}_d = 0$ here, so its Ω_d contains only the yaw-rate term.

b Tilt-cone limit `control/shaping.py`, vertical plane through F



Limited_thrust_vector: first scale a to $\|a\| \leq a_{\max}$ (if set), then $F = m(a - g)$; outside the cone it is replaced by its orthogonal projection $\max(0, \sin \theta \|F_{xy}\| + \cos \theta F_z) (\sin \theta \hat{F}_{xy}, \cos \theta)$, including the apex. A limited step rolls back integral growth (anti-windup).

c Smooth SO(3) error: $\|e_R\| = \sin \varphi$



$$\begin{aligned} e_R &= \frac{1}{2} (R_d^T R - R^T R_d)^\vee \\ &= \sin \varphi \hat{k} \text{ for } R_d^T R = e^{\varphi \hat{k}}; \\ &\text{restoring torque vanishes} \\ &\text{at exactly } 180^\circ, \text{ so the} \\ &\text{controllers target upright} \\ &\text{flight (tilt} < \pi/2), \text{ not} \\ &\text{inverted recovery.} \end{aligned}$$